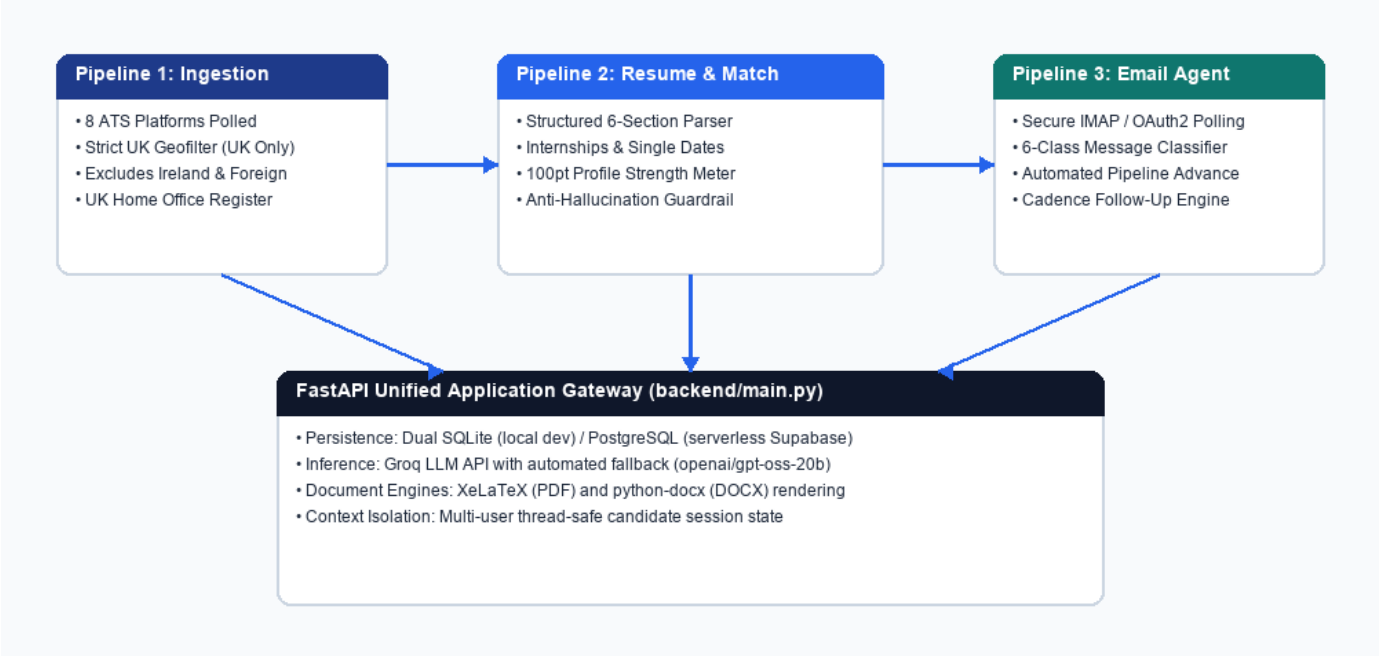
Cafy Agent - Enterprise Platform Specifications

Production Architecture, Autonomous Pipelines, and Complete UI Showcase

1. Executive System Architecture

Cafy Agent is an enterprise-grade ATS automation and career intelligence platform designed for the UK market. Built on a high-throughput FastAPI backend and a responsive React Vite frontend, the system unifies three autonomous pipelines: Pipeline 1 continuously ingests and geofilters vacancies strictly within the United Kingdom; Pipeline 2 delivers structured resume parsing, 100-point profile strength evaluation, hybrid job matching, and zero-hallucination AI resume tailoring; and Pipeline 3 operates an autonomous Email Agent that synchronizes recruiter correspondence and automates pipeline status transitions.

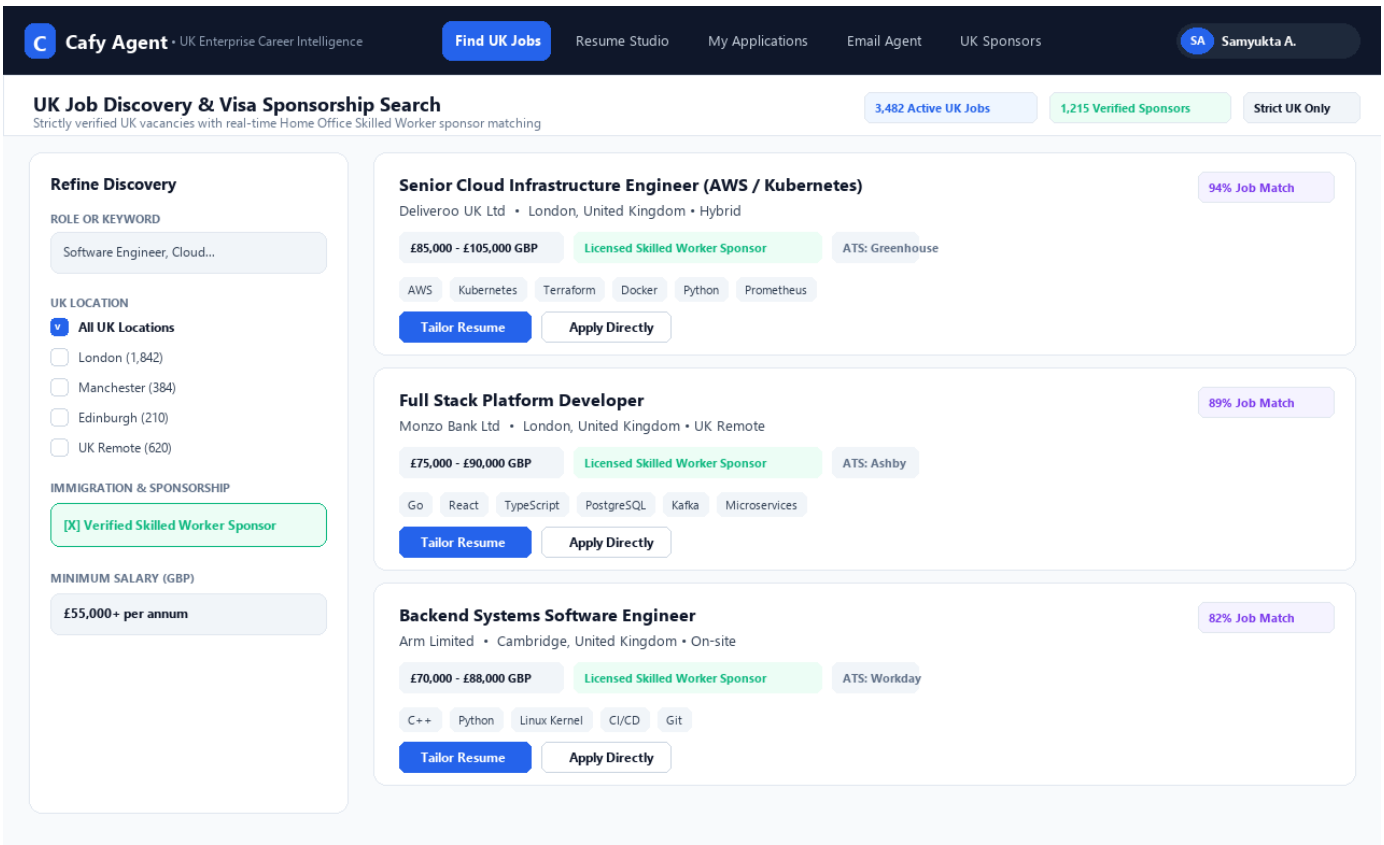


2. Core Engineering Pillars

• Strict UK Containment: Employs negative regex to reject overseas and Irish postings, coupled with positive matching for 25+ UK hubs.
• Home Office Sponsorship Integration: Verifies employers against the UK Home Office Worker Register (Skilled Worker sponsor list).
• Factual Invariance & Factual Guardrails: Prevents LLM hallucinations using strict token and metric verification.
• Production Dual Compilers: Generates publication-grade XeLaTeX single-page PDFs and formatted Microsoft Word (.docx) files.

3. Feature 1: UK Job Discovery & Visa Sponsorship Search Engine

The job discovery engine aggregates listings across 8 enterprise ATS systems (Greenhouse, Lever, Ashby, Workday, Workable, Oracle Cloud, BreezyHR, and Pinpoint). Workday network faceting filters foreign listings before download, while the normalizer strictly validates UK locations. Every role is cross-referenced against the UK Home Office Worker Register to badge licensed Skilled Worker sponsors.

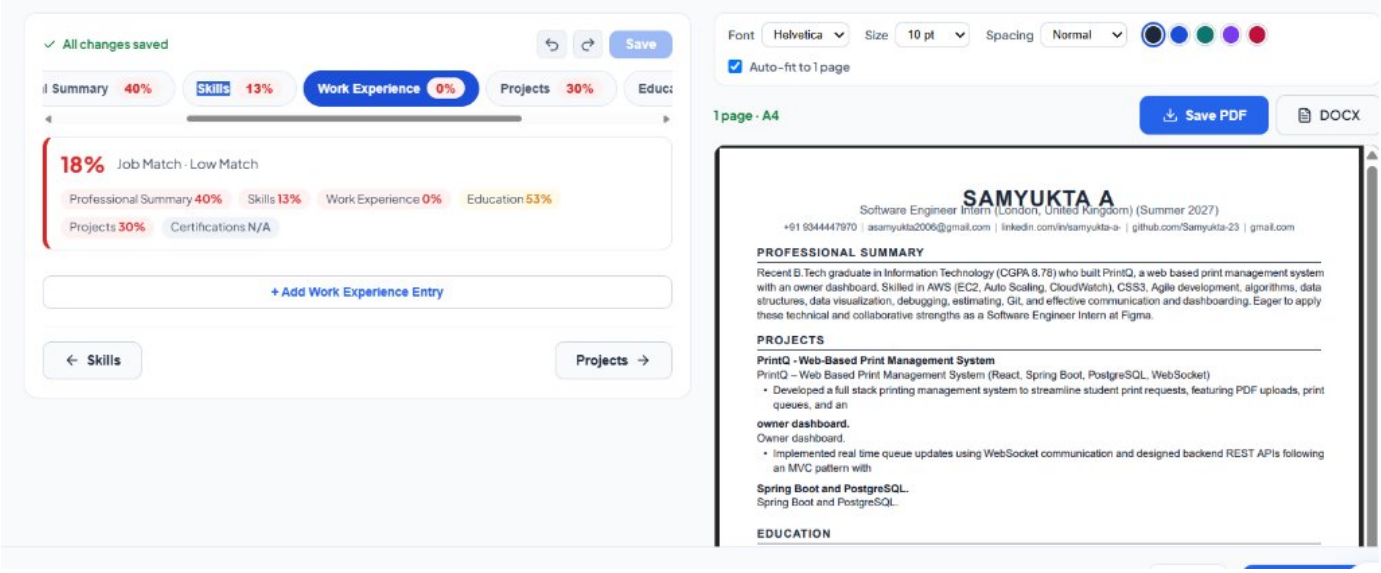


Key Interface Capabilities:

- Municipal Corridor Filtering: Filter by London, Manchester, Edinburgh, Bristol, Cambridge, or UK-wide remote.
- Immigration & Sponsorship Badges: Instantly isolate verified Skilled Worker visa sponsors from standard listings.
- Normalized Compensation: All salaries standardized to annual GBP ranges with minimum threshold filters.

4. Feature 2: Resume Studio, Match Scoring & Live PDF Workspace

The Resume Studio provides structured 6-section profile extraction, real-time job match scoring, and instant XeLaTeX compilation. The parser seamlessly extracts single-term dates, date ranges, and internship experiences, crediting full experience to candidate profiles.



Key Interface Capabilities:

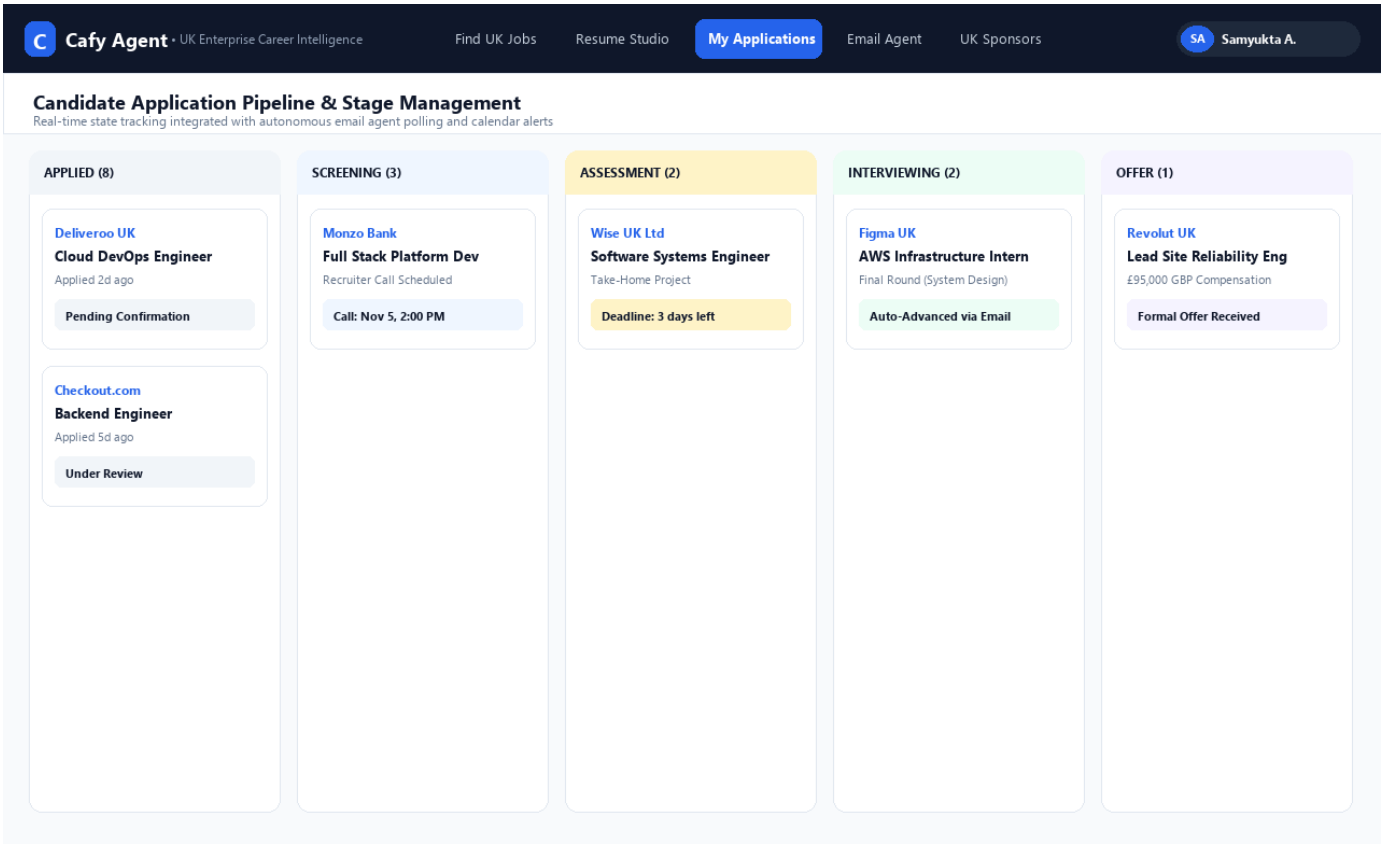
• Live Section Match Meters: Displays real-time fit across Professional Summary, Skills, Work History, Education, and Projects.

• Factual Action Rewriting: Utilizes Groq LLM inference with automated fallback and zero-hallucination verification.

• Instant Multi-Format Downloads: One-click export to publication-ready single-page PDF (XeLaTeX) and editable Word document (.docx).

5. Feature 3: Candidate Application Pipeline & Stage Management

Candidates manage their active application funnel using an interactive, state-synchronized pipeline. Applications advance through stages (Applied -> Screening -> Assessment -> Interviewing -> Offer) via candidate actions or automated email notifications.



Key Interface Capabilities:

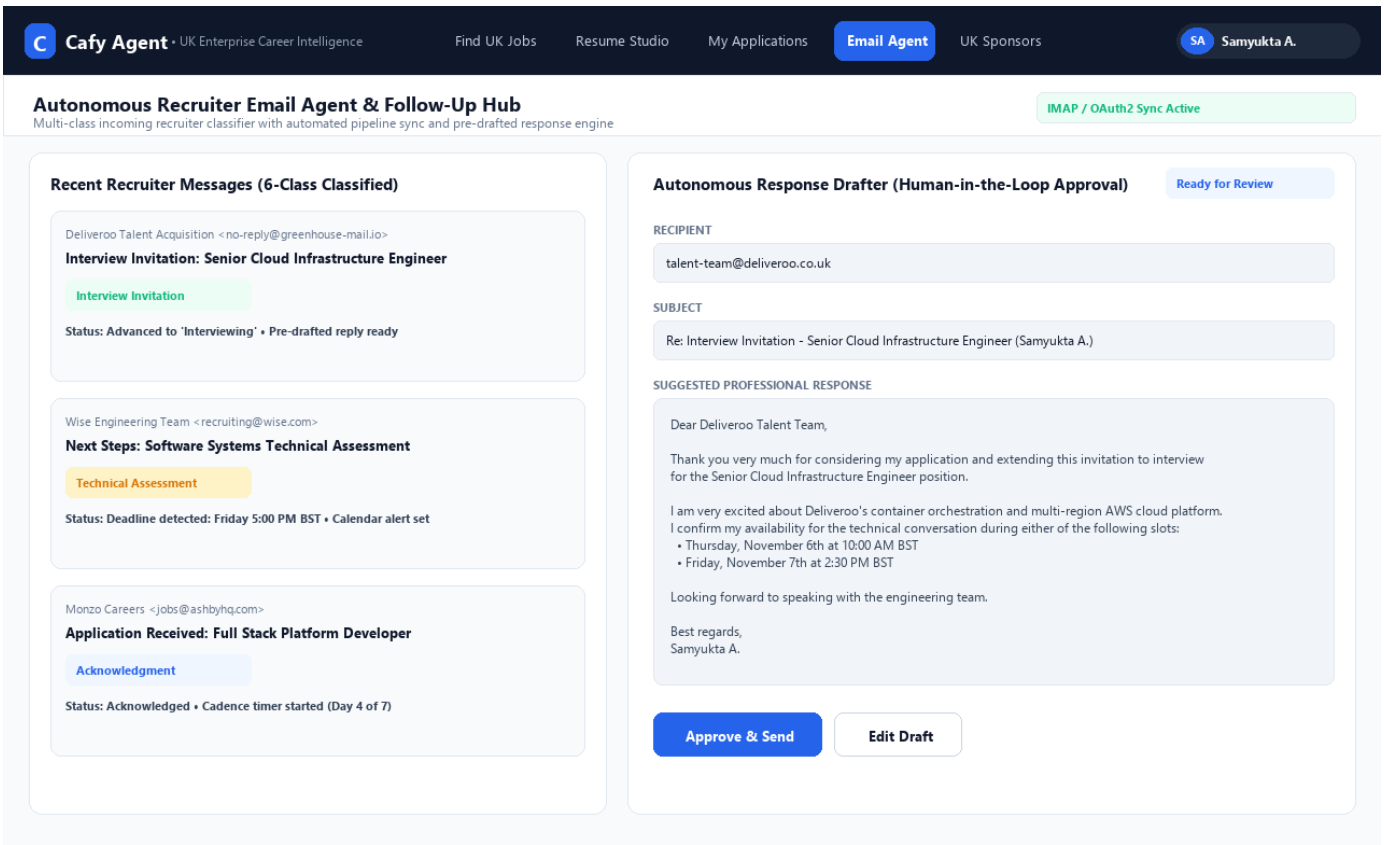
• Multi-Stage Lifecycle Board: Instant visibility into pending recruiter calls, technical assessments, and formal offers.

• Deadline & Calendar Alerts: Extracts submission deadlines for take-home coding challenges with visual countdown badges.

• Automated State Progression: Auto-promotes candidates to 'Interviewing' immediately upon receiving recruiter scheduling links.

6. Feature 4: Autonomous Recruiter Email Agent & Follow-Up Hub

Pipeline 3 connects candidate inboxes via secure IMAP or OAuth2 protocols. Recruiter emails and ATS notification relays (@greenhouse-mail.io, @ashbyhq.com) are classified by a 6-class neural categorizer, updating application states and pre-drafting professional responses.



Key Interface Capabilities:

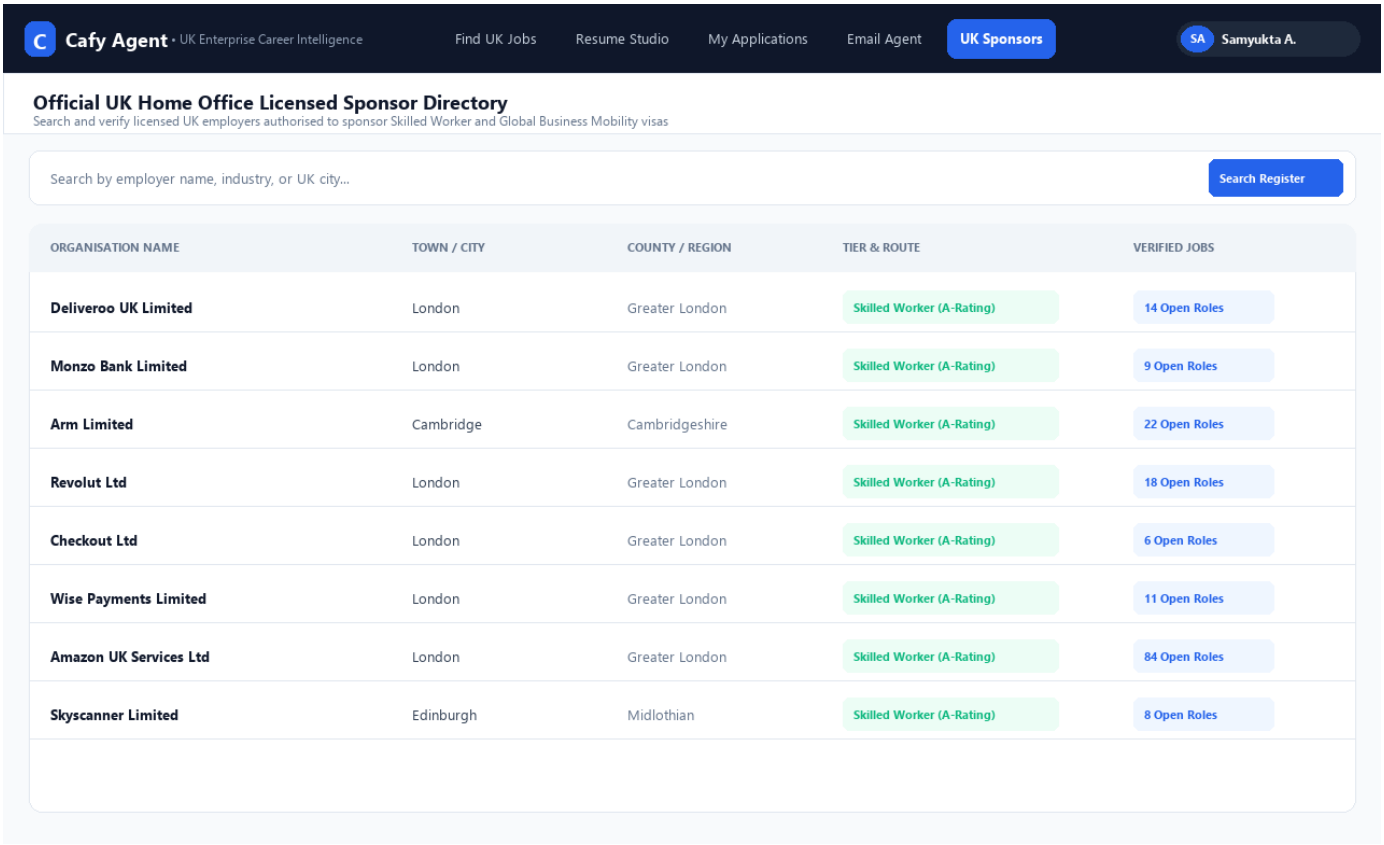
• 6-Class Categorization Stream: Distinguishes Interview Invitations, Technical Assessments, Acknowledgments, Offers, and Rejections.

• Human-in-the-Loop Approval: Pre-drafts context-aware acceptance and availability replies for one-click candidate review.

• Follow-Up Cadence Automation: Detects applications with no recruiter reply after 7 to 10 business days and queues inquiry drafts.

7. Feature 5: Official UK Home Office Licensed Sponsor Directory

An authoritative directory of licensed UK employers authorised to sponsor international professionals under the Skilled Worker route. Cross-references the official UK Home Office Register of Licensed Sponsors with city and county verification.



Key Interface Capabilities:

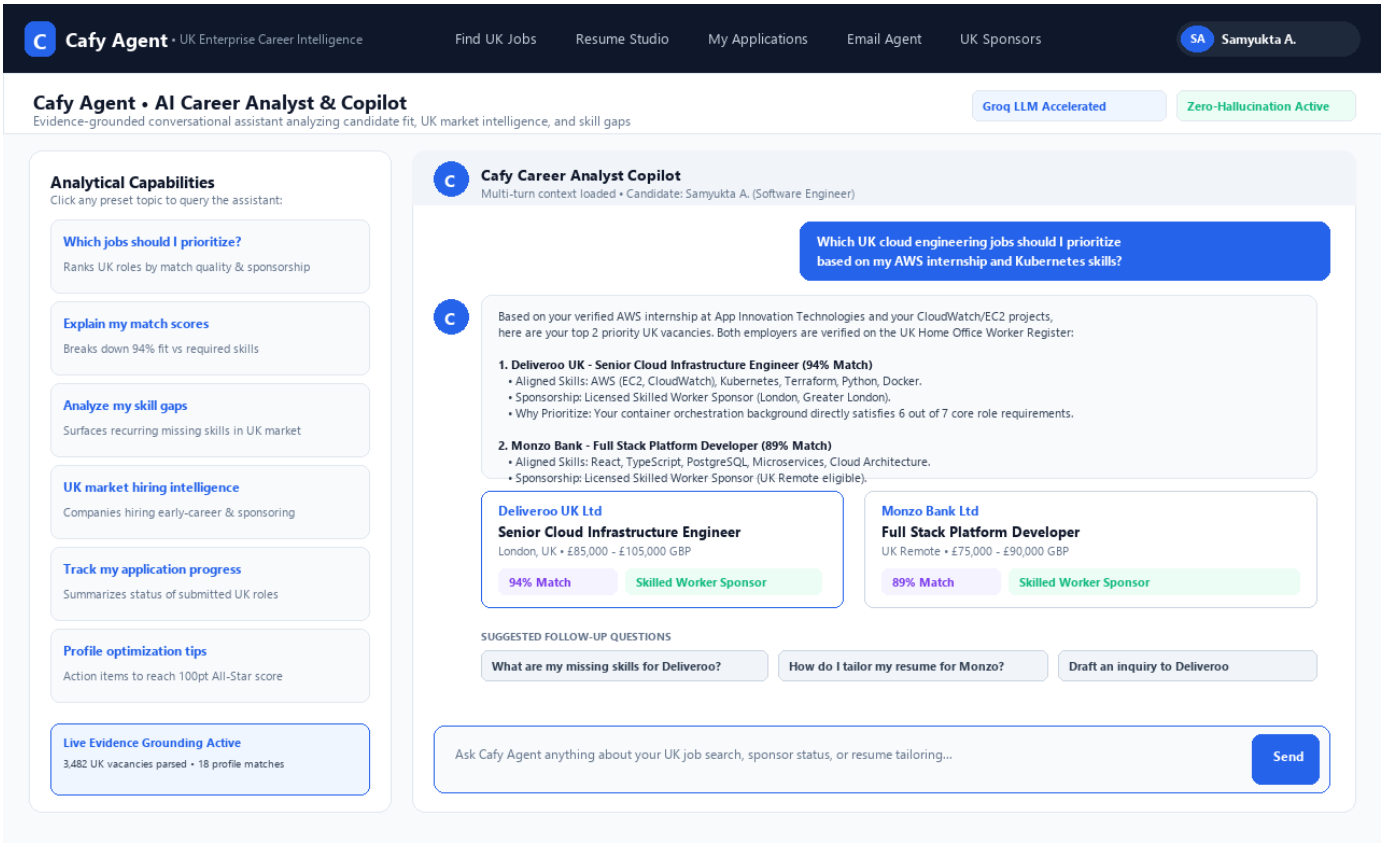
• Live Register Search: Search by corporate legal entity, trading name, industry, or UK town/county.

• Licensing Tier Verification: Highlights Skilled Worker (A-Rating), Global Business Mobility, and Scale-up routes.

• Real-Time Vacancy Cross-Reference: Connects each licensed sponsor directly to live open vacancies discovered across ATS platforms.

8. Feature 6: Cafy Agent - AI Career Analyst & Copilot (Chatbot)

An evidence-grounded conversational career analyst accessible through an interactive slide-out dock. Powered by Groq LLM inference, the chatbot analyzes candidate resume facts against active UK job listings to answer strategic application questions in real time.



Key Interface Capabilities:

• Strategic Application Prioritization: Recommends which UK roles to apply to first based on match score, recency, and sponsorship viability.

• Match Fit Explanation: Granularly explains why a candidate received a 94% or 82% fit, isolating matched vs. missing technical skills.

• Market-Driven Skill Gap Analysis: Highlights recurring missing competencies across current UK vacancies with high-ROI learning roadmaps.

• Interactive In-Chat Job Cards: Renders real-time, clickable recommendation cards with direct 'Tailor Resume' action buttons.

• Strict Zero-Hallucination Bound: Constrained to real database evidence; prohibited from inventing scores, employers, or qualifications.